07.08.2005 FEATURE ARTICLE

The African Hacker

By Pascal G. Zachary

07.08.2005 LISTEN

With home-brewed code and a little help from Microsoft, a programmer in Ghana launches Africa's first software empire By G. Pascal Zachary Photography by Eric Don-Arthur The first time I meet Hermann Chinery-Hesse, he is pouring diesel fuel from a plastic jug into an electric generator. I am in the West African country of Ghana, visiting his software company, Soft Tribe Ltd. Chinery-Hesse is chief of the Tribe. He's made a small fortune writing software, working as a systems architect, and selling computer code to hundreds of businesses in his country of 21 million people. He drives a Mercedes. He wears imported Birkenstock sandals. He hails from a prominent family, was born in Dublin, and went to college in the United States. He could be working anywhere on the strength of his Irish passport, yet he's spent the past dozen years in Accra, Ghana's coastal capital and one-time slave-trade hub. This steamy December morning, with deadlines looming, his electricity is out, his programmers are idle, and he's feeding fuel to a balky 50-kilovolt-ampere generator—one of the three he keeps at the ready. Having emptied his container and thus delivered power to his 18 programmers—about one-tenth of all full-time code writers in Ghana—Chinery-Hesse relaxes and, for the first time, acknowledges my presence. Stroking his beard, he quips, "If we Africans are to develop, we must want to get our hands dirty." Chinery-Hesse thinks a lot about how Africans can better their lives. And he's not alone in his conviction that his people can thrive by harnessing innovations in computing, electronics, and telecommunications. "Technology," he declares, "is on our side." What sets him apart, other than a knack for adapting code to Africa's distinctive conditions, is his willingness to embody his beliefs in a business enterprise. With 65 employees, his company, popularly known as "Soft," is a testimony to the idea that information technology can be a great equalizer for his people. The company's software—written in C++, Visual Basic, and even the ancient Clipper development language for DOS—is everywhere in Accra. Chinery-Hesse, now 41 years old, started out pursuing a "shotgun business strategy," as he puts it, offering between 15 and 30 programs at any given time, including his best seller, a payroll program called Akatua. In contrast, mighty Adobe Systems Inc., in San Jose, Calif., with nearly 4000 employees, offers about 45 software products. With Soft's abundance of offerings, the company has grown at an impressive clip over the past decade, doubling in revenue and staff year after year. Old, cheap PCs are the backbone of African offices and increasingly popular cybercafés, where ordinary sub-Saharan Africans, most of whom subsist on less than US $1 per day, access the Internet. The people of Ghana, particularly those in Accra, a sprawling city of 2 million, are a

little better off than their sub-Saharan neighbors, earning on average about $2 per day in an economy based primarily on exports of gold, timber, and cocoa. Slightly smaller in area than the state of Oregon and sandwiched between Togo and Ivory Coast on the Atlantic Ocean, Ghana won independence from Britain in 1957, the first black African nation to do so. In 1992, following a series of military coups and dictatorships, Ghana adopted a new constitution that provided for a democratic political system. Lt. Jerry Rawlings, who had ruled the country as a dictator from 1981 to 1992 won the first two elections in 1992 and 1996. Rawlings peacefully transferred power to the current president, John Kufuor, in 2000. Political power struggles, civil war, corruption, disease, and wasted potential usually spring to mind when the subject is Africa. But I know a different Africa, from many visits in recent years, and my different Africa springs in part from my contact with Chinery-Hesse. Much of his success reflects his shrewd realization that Africans can't simply import American or European software programs but need to tailor them to their own surroundings. Chinery-Hesse's ability to build a thriving software business in an economic climate where the need for information technology is just beginning to grow has tested his ingenuity and made him a celebrity in Ghana [see photo, "Making a Splash"]. While hardly in the global technical vanguard, Chinery-Hesse is counted as one of "a handful of the most important software developers in Africa," says Eric Osiakwan, an IT specialist in Accra who consults for the World Bank. "Hermann is our Bill Gates, and Soft is our national software champion." Chinery-Hesse is thus a valuable role model. "Many young code writers get their start at Soft, learn the ropes, and then hone their skills," says Guido Sohne—himself an example of Chinery-Hesse's influence: a former employee of Soft, he now runs his own business as an independent programmer in Ghana. "To us, at least, Hermann is a savior," Sohne says. "Saviors" are common in Africa, and they usually come in the form of demagogues or rebel leaders, missionaries or medical doctors, peacekeepers or refugee-camp managers. Rarely are they Birkenstock-wearing engineers or software programmers. That's why Chinery-Hesse is worth getting to know. Because if Africa has a sunny future, Chinery-Hesse will be a part of it. He is emblematic of the little-known world of African code writers, hackers, engineers, and entrepreneurs who have chosen to live and work in their homelands, persevering against great odds and on the margins of global technological change. These people, often self-taught and sometimes surprisingly well paid, are the beneficiaries of the accelerating spread of the Internet, the increasing power of cheap computers, and the burgeoning global community of programmers. The African hackers are quiet heroes, however, because they embody a side of the region that is entirely missed by the world's media: they represent an Africa where blacks are using their brains to try to build a better future against a backdrop of spotty electricity, unfettered piracy, inferior computers, and mediocre universities. Nowadays, Soft is focusing on document management software for government agencies and large multinationals doing business in Ghana. It also supplies accounting software for small and medium-size businesses, such as stores and restaurants. Its cash register program, for instance, tallies the bill at Accra's largest grocery store. And one of the most popular dining spots in Accra, Frankie's Hotel and Restaurant, uses Soft's code to manage inventory, process payroll, and pay taxes [see photo, "Would You Like Code With That?"]. Hotels, gold-mining companies, and Internet cafés do, too. Even a few cocoa plantations calculate their piecework pay with Soft's code. Here's why Soft's programs are essential: programs from the United States and Europe are usually too expensive and require lots of memory and the latest machines. By contrast, those that Soft offers are small enough to work with the Intel 486- and even 386-based PCs that are readily available in Ghana for as little as $100. CHINERY-HESSE CALLS HIS AFRICANIZED CODE "tropically tolerant." Besides being compact, his programs also write frequently to disk, reducing the chances of losing data if power is lost, as it often is. Because Internet connectivity remains relatively expensive, his programs also work offline as much as possible. And to combat the rampant piracy, beta versions of software rarely leave Soft's premises, finished products don't have an autoinstall function (you need a Soft technician to launch them), and batches of bug fixes are often delivered individually to customers rather than generally released. Why doesn't Soft write everything in Linux, the open-source language available for little or no cost? It would seem a no-brainer for a software entrepreneur in a desperately poor region of the world. Not so, and the reasons show why Chinery-Hesse is no ordinary third-world software tycoon. Linux is simply impractical for his purposes. "It can't be used for serious business," he says, because it would be too easy for employees of a business to learn to use—and abuse—the source code of essential programs. "In Accra, an IT manager earns $100 a month," Chinery-Hesse explains. "He has access to all of a company's data, including backups. With open-

source, he can learn to generate bogus reports. He'll delete charges and pay himself or others money." Besides, "if we went for open-source, we would be relegated to basically doing installation and training," he adds. "Do I want a country of software developers or a country of installers?" Chinery-Hesse's answer can be deduced from watching him ride along one of Accra's traffic-clogged roads on a motorbike, a risky form of travel he undertakes because, as a driver in a hurry, he can more easily dodge potholes and lines of cars. The trip leaves him blanketed with dust from the harmattan, the winds that carry sand from the Sahara, thousands of miles away, into Accra. His willingness to brave both the harmattan winds and Accra's notorious traffic jams underscores Chinery-Hesse's ambition. Despite the disadvantages of his country's geography and infrastructure, he wants Ghana—and Africa generally—to achieve a kind of greatness in software, an indigenous engineering success that has perpetually eluded Africans. The challenge is both technical and social, because in Africa the business of writing code is tightly coupled with the economics of underdevelopment. Soft's local market is tiny by international standards, worth only millions of dollars annually, and the public appetite for innovations is limited. "The big problem in Ghana is that Soft can do really cool, complex stuff, but there's not a whole lot of people who want that stuff," says former Soft programmer Sohne. "They might need it, but they don't know they need it." "Hermann is our Bill Gates, and Soft is our national software champion" For instance, Sohne could design an automatic bill payment system for the electric company, whose customers must still go to an office to pay their bills. But because the company is a state-owned monopoly, it has no incentive to make payment more convenient. The lack of daring among African IT customers forces code writers in the region to look elsewhere for ideas and energy. To expand the company's business both within and beyond Ghana's borders, this past April Chinery-Hesse struck a deal with Microsoft Corp., Redmond, Wash., that for the first time has him selling software made by another company. He's agreed to peddle a suite of applications, called Navision, aimed at medium-size businesses. It coordinates a company's financials, supply chain, customer information, and Web-based transactions through a single database. The deal gives Chinery-Hesse rights to the source code, so his programmers can produce add-on programs, including ones that manage payroll and microfinancial transactions, such as small loans to people too poor to qualify for traditional bank loans. The agreement lets Soft sell its add-on software—and also allows Microsoft to market Soft's add-ons throughout the world. "We deliver the building blocks, and Soft brings local knowledge," says Microsoft's Andreas Peterson, a Navision product manager based in Nairobi, Kenya. "Hermann can take what we've done, customize it, and sell it internationally." Microsoft has 1500 developers working on Navision worldwide but only a handful in tropical Africa, including those in Kenya as well as in Nigeria. "The recognition of Soft Tribe by Microsoft is a source of great pride to the whole country," the Ghanaian president, Kufuor, said at the announcement of the deal in April. SOFTWARE VENDORS doing business in Africa constantly battle technological ignorance. Take Soft's latest large assignment, for example. It's a $200 000 contract to create a software program to manage all case documents for the government of Ghana's human rights commission, which investigates violations against individual citizens. When Chinery-Hesse bid on the contract, which is being funded by a Danish aid agency, he based his proposal on the belief that the rights commission had an existing computer network linking its dozen field offices to its headquarters in the capital. Only after Soft won the contract did Chinery-Hesse realize that there was no such network. Such misunderstandings happen often. Sometimes, Soft misjudges the client's capacities, but this time the client misjudged its own needs. Because all of the rights documents are currently done on paper, there wasn't anyone at the commission who knew enough about computing to realize that the commission needed not only a program for creating and managing documents but also a network for sharing them. Says Chinery-Hesse, "Customers often can't see the difference between a program and a network." So Soft is building both, applying the tropically tolerant mind-set to the design. This being Ghana, a country whose fully democratic government is only five years old, Chinery-Hesse is designing a Microsoft Windows program that will safeguard rights information in the event of, say, a military coup or a physical attack on the commission's offices. Relying on a "store and forward" technique, Soft's program calls for every case file to be stored in a central server, most likely to be located in London. Nor will the program depend on constant Web access. Instead, it will store files locally on a computer, shipping them across the network automatically whenever a connection is made. "This way, if the Web connection goes down, as it surely will, work continues," Chinery-Hesse explains. In the

same way, if the computers in a commission office are stolen or broken, the server can send copies of the office's entire files as soon as the computers are replaced. Understanding local conditions and designing software to suit them are what Chinery-Hesse does best. Consider his experience with Internet cafés. In Ghana, as in most of the developing world, Internet service remains expensive. Because there are very few landlines, dial-up service from the home or office is virtually nonexistent. The next best option, a very-small-aperture terminal satellite (VSAT) connection, starts at about $5000 for the dish antenna, modem, router, and cabling, prohibitively expensive for the average Ghanaian. So most people pay $1 per hour to connect to the Web from one of the hundreds of cafés outfitted with satellite dishes and low-end PCs that have sprung up in the country over just the last five years. Each café needs software to manage its operations and, most important, to clock the time a customer spends surfing the Web. Losing track of time means losing money. Because the large U.S. and European software companies don't specialize in such software, Chinery-Hesse seized the opportunity to satisfy a large local need. What happened next, however, reveals the vagaries of writing software in one of the world's poorest places. Chinery-Hesse won a contract with the largest Web café in Ghana to develop a basic program, called Lumumbashi (after a town in the Democratic Republic of Congo). Lumumbashi's chief feature was to track usage while preventing wrongdoers from turning off the feature and surfing for free. Whenever possible, Chinery-Hesse takes his custom-made programs and converts them into generic ones for other jobs. Following this model, he developed a general piece of software for Web cafés based on Lumumbashi. Anticipating threats from software pirates, he sold test versions to a first batch of customers and then offered a $50 prize to users who crashed his system and could explain to him how they did it. The feedback helped make Lumumbashi essentially immune to tampering. But Soft failed to sell many copies for a reason Chinery-Hesse never anticipated: there were so many pirated copies of other café programs that few managers needed to purchase Soft's. BESIDES PROVIDING GHANA with tropically tolerant code, Soft is the premier training ground for the country's growing population of full-time programmers, who today number around 200. Their life isn't easy. On a humid January morning, Rudolf Akrong is one of seven male programmers crammed into a tiny windowless room. Ancient PCs line three walls; along the fourth, an administrator sits at a desk elevated on a small platform [see photo, "Soft's HQ"]. Akrong is scrambling to make a change in an accounting program used by Frankie's, which happens to be one of Soft's most important customers. Frankie's owner wants a new feature in its customer receipts, a line that separately lists the sales tax imposed on each transaction; currently only the total amount is given. Chinery-Hesse asked for the change to be made in a single day, a demand Akrong is finding impossible to meet. He slept at the office last night on a small foam pad, and this morning, with the electricity out, he is working on his laptop while other programmers, who have only desktop PCs, lounge about, cursing and joking about the electric company and waiting for Chinery-Hesse to fire up the backup generator. The laptop displays a photo, in the background, of Akrong's girlfriend, who is attending the government university in Accra. By midmorning, Akrong still hasn't fixed the bug. He fears igniting Chinery-Hesse's wrath, but his concentration is interrupted by a friend's arrival. The friend is showing off a new PDA from Dell Computer Corp., Round Rock, Texas. It's no small event: the chance to handle the latest electronic gear from the United States or Japan is rare. The programmers gather around, chatting excitedly in Twi, the local language. Akrong gamely tries to keep working, but after a half hour, he pushes away from his desk and presses his hands against the sides of his head. A colleague suggests lunch, and he agrees. The programmers pile into Akrong's 13-year-old Peugeot 405. Alone among the programmers, he owns a car, a bonus from Chinery-Hesse. Fortunately, it starts without a push (it doesn't always). On the drive to a nearby restaurant, the programmers joke about girls, music, and the PDA. At the restaurant, they are treated with respect. Though they rank on the bottom rung of the software pecking order globally, in Ghana their salaries put them squarely in the middle class, and without families, they freely spend their cash. The waitresses jump to attention on their arrival at the restaurant. The programmers order a special local dish called waakye, a hefty plate of rice and beans drenched in a spicy, oily sauce. As a side dish, a local bony fish, tilapia, is served. The feast costs $4 a person, a dear price in Ghana, but no sweat for programmers like Akrong. He earns about $400 a month at Soft—about five times as much as an experienced Ghanaian nurse earns and 10 times as much as a policeman earns. Back at Soft's office, Akrong returns to his bug fix. He plays jazz on his laptop and soon dozes off. Sleeping in the office the night before—or maybe the heavy lunch—is catching up with him. Akrong, 23, has worked at Soft for the past four years. After high school, he attended a one-year course on

basic science, then was hired by Chinery-Hesse. Accra is home to the country's flagship public university, but Chinery-Hesse doesn't recruit its graduates, finding the university's computer science program woefully out of date. Assembly languages from the 1960s are still taught religiously at the university, for instance, but courses on such contemporary languages as C++ are offered only sporadically. The computer science department's dozen working PCs are not generally connected to the Internet. Some students write their programming exercises with pencil and paper, while others work in Internet cafés. The department has only three permanent faculty members to serve 400 students. The department chair, who has a master's degree, earns about $300 a month. Academic salaries are so low that students are advised to leave the classroom whenever their teacher is 10 minutes late; odds are the professor is consulting somewhere and won't show up at all. Because of the spotty quality of formal computer training available in the country, Chinery-Hesse prefers to hire what he calls "uncorrupted" minds that he can mold. "Ghana's educational system is exceedingly theoretical and designed to train people to manage, say, a Mercedes dealership rather than to build a Mercedes," Chinery-Hesse says. "What we require is more practical education." Chinery-Hesse offers such an education for those persistent enough to stick it out. Akrong, for instance, began as an unpaid intern. "Then, the simplest thing I couldn't handle," he says. "They were hard on me. There was a culture of certain people feeling superior, who sought to put you down. I got dissed." The trial by fire was worth it. Today he is one of Soft's most important product tweakers and bug fixers. He writes applications for Windows programs in C++ and Visual Basic, and he even has a growing entourage of junior programmers who are learning at his knee. The day he finishes the code for the restaurant program, Chinery-Hesse tells him he'll be training a new Soft employee, whom he'll help choose. He is told to evaluate the C++ skills of the various candidates and to quiz them on their business know-how. Training new staff is a constant at Soft, where programmer turnover is astonishingly high. Some can't hack the regimen, while others strike out on their own, seeking higher pay and more autonomy. Chinery-Hesse reserves his special pique for those who leave to seek programming jobs in the United States or Europe. "We should fix what we've got in Ghana," he says. Akrong wants to remain in Ghana, but he realizes his code writing may suffer if he does. There are few places for him to study formally, so he learns on the fly. He tries to stay sharp by browsing the Web and rapping with his father, who years ago worked as a programmer at Soft. Chinery-Hesse offered to teach Akrong the ropes as a favor to his dad. FAMILY TIES MEAN A LOT in Africa. Chinery-Hesse's mother was once the highest-ranking African woman in the United Nations system, rising to the post of deputy director of the International Labor Organization before retiring in 1999. His father, whose ancestors include a German colonist who bequeathed the Hesse name, is a lawyer who advises the government of Uganda. After spending his early years outside of Ghana, Chinery-Hesse attended high school in the country and then enrolled at Texas State University, in San Marcos, Texas, where he lived with a relative. His experience in the United States transformed him. "I could suddenly see opportunities more clearly in Ghana after being in America," he says. Before his U.S. stretch, he dreamed of becoming a civil servant, following his mother's path, and he blames Ghana's educational system for such personal conservatism. "We aren't trained to be creative," he says. America taught him the value of improvisation, and that talent is "not a matter of qualifications but of skills." He also discovered an ability to bounce back from rejection. "America taught me to accept 'no' and to move on." After graduating with a bachelor's degree in industrial technology, Chinery-Hesse moved to London, working for a couple of years in business before coming to Accra in a fit of idealism in 1990. The country, once one of Africa's richest, had seen its economy collapse under the stress of repeated military coups. Unfazed, Chinery-Hesse quickly persuaded the leading travel agent in Accra to let him computerize his back office by creating a program to keep track of expenses paid and of clients who owed him money. Chinery-Hesse aced the job and then recruited a mate from secondary school, Joe Jackson, who brought important management skills and market knowledge to the venture. "My parents' financial comfort allowed me to gamble," Chinery-Hesse says. His bet paid off. Ghana was starting to computerize, and even many of the country's largest businesses found importing European computer consultants to be too expensive. Chinery-Hesse positioned Soft to fill the void. At first, locals who doubted his knowledge of computing dismissed Chinery-Hesse. But over time, he wore down his skeptics. Heineken, Nestlé, and Unilever—three European brands with major operations in Ghana—became large customers of business applications, choosing Soft for the same reasons local

companies would: its programs were cheaper, they were more suited to the old computers in use in Ghana, and they were customized to local business conditions. Success convinced Chinery-Hesse that "growing his own" code was essential to building both a business and a better society. But skepticism remains. Multinational corporations, while among his best customers, constantly question whether they should be buying African-made software even for use in Africa. Meanwhile, the few large, lucrative government software projects that come up are routinely shipped abroad, handled by expensive European or U.S. programmers. Even so, Soft has inspired imitators in recent years. For example, a former Oracle manager, a native of Ghana, has launched a firm. So have other engineers educated in the United States. Talented programmers from other African countries also have trickled into Ghana, drawn by the country's tranquility, ease, and low cost of living. "We are all doing the same thing and chasing the same 10 customers," says Mauli Tse, a Ghanaian graduate of the Massachusetts Institute of Technology, in Cambridge, who runs an IT business in Accra. A few coding shops in Ghana are chasing outsourcing jobs from the developed world as well, but not Chinery-Hesse's. He prefers to pursue customers in neighboring African countries, where his philosophy of tropically tolerant code matches local conditions. Lately, Chinery-Hesse is reducing the number of products the company offers to focus on its strengths in accounting and document management software and to concentrate his programmers on making add-on products for Navision. Chinery-Hesse and his backers know the Microsoft deal is Soft's chance to succeed internationally with a made-in-Ghana software program that piggybacks on Microsoft's marketing muscle. Committed to self-reliance, Chinery-Hesse thinks that his fellow Africans must accept that, while homegrown information technology may save them, the wider world won't. His dream remains "to expand software in Ghana, to make more money, and to show the world" that, despite the often depressing headlines about poverty and mayhem in Africa, "there are opportunities here for technical talent." He has personally reaped the benefits from these opportunities. The question in his mind is: how many more Africans will, and how soon? With additional reporting by Daniel Morris ABOUT THE AUTHOR G. Pascal Zachary is the author of The Diversity Advantage: Multicultural Identity in the New World Economy (2003). His research on information technology in Africa was supported in part by the Center for Science, Policy and Outcomes at Arizona State University, in Tempe.

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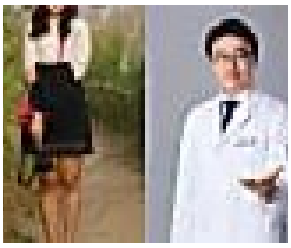

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